Highlights

TRACE-SL: Designing Traffic Sensors for Recoverability

Anonymous Author

- We formulate sparse traffic sensing as reconstruction-aware hidden-network design.
- TRACE-SL combines posterior certificates, validation selection, and local swap refinement.
- Held-out Stage12 evidence spans PeMS7_228, PeMS7_1026, and Seattle.
- Submission-facing claims are explicitly bounded by evidence and scoped theory.

TRACE-SL: Designing Traffic Sensors for Recoverability

Anonymous Author[a]

[a]Anonymous Institution, Anonymous City, Anonymous Country

ARTICLE INFO

Keywords:
traffic sensor placement
network design
traffic state reconstruction
optimal experimental design
graph signal sampling

ABSTRACT

Sparse traffic sensor deployments are long-lived infrastructure decisions: once a subset of links or stations is instrumented, the remaining network state must be inferred from partial observations. We reformulate sparse traffic sensor placement as an inverse-design problem: the goal is not to maximize coverage, variance, or topological visibility at observed locations alone, but to choose sensors that make the unobserved network state recoverable under a transparent reconstruction model. We propose TRACE-SL, a reconstruction-aware layout framework that couples GLS/MAP and graph-signal reconstruction with posterior uncertainty certificates, validation-calibrated score selection, and validation-aware local swap refinement. The design protocol keeps model fitting, layout selection, and held-out evaluation separate. On PeMS7_228 Stage12 evidence, TRACE-SL reduces held-out GLS/MAP MAE relative to validation-selected random layouts by 0.123, 0.095, and 0.163 at 10%, 20%, and 30% sensor budgets, respectively. PeMS7_1026 and Seattle provide additional ten-split baseline comparisons, including low-budget cases where internal A-trace or multistart variants are competitive. The contribution is not a claim of globally optimal or universally robust placement; it is a transparent network-design workflow whose claims are tied to formal scoped properties, held-out evidence, and auditable result artifacts.

1. Introduction

Traffic agencies rarely observe a road network completely. Fixed loop detectors, cameras, probe feeds, and connected-vehicle observations provide partial measurements, while the operational question often concerns the full network state. The siting decision is therefore not only a coverage problem. A sensor can be valuable because it makes unobserved links recoverable under a transparent estimation model, even when it is not the highest-flow or highest-variance location.

This paper studies sparse traffic sensor layout as reconstruction-aware network design. Given a fixed budget, we choose a set of candidate nodes or links to observe and then evaluate how well a transparent estimator reconstructs the hidden complement. This differs from classical count-location and observability formulations that emphasize origin-destination coverage, flow interception, link independence, or full link-flow inference (Yang and Zhou, 1998; Gentili and Mirchandani, 2011; Hu, Peeta and Chu, 2009; He, 2013); it also differs from traffic state estimation and forecasting work that improves estimators while taking the observation pattern as fixed (Coric, Djuric and Vucetic, 2012; Seo, Bayen, Kusakabe and Asakura, 2017; Li, Yu, Shahabi and Liu, 2018; Yu, Yin and Zhu, 2018). The layout and the reconstruction model must be evaluated together.

We propose TRACE-SL, a transparent reconstruction-aware sensor layout framework. TRACE-SL builds candidate layouts using posterior trace, log-determinant, scenario risk, coverage, and low-dimensional reconstruction heuristics; selects among candidate scoring rules by validation reconstruction error; and refines selected candidates with a validation-aware one-swap search. The final evaluation is held out: once the layout rule is fixed, only selected sensors are observed and non-sensor nodes are reconstructed and scored.

The contributions are fourfold. First, we formulate fixed-budget sensor layout for hidden-network reconstruction with explicit train, selector-validation, tuner-validation, and test separation. Second, we give formal scoped properties for the transparent design layer: the posterior trace identity under a GLS/MAP model, covariance monotonicity for fixed targets, a finite-candidate validation selection bound, and one-swap finite termination/local optimality. Third, we provide Stage12 held-out evidence against random, validation-selected random, variance, observability, graph sampling, A/D-optimal, and POD-style baselines across PeMS7_228, PeMS7_1026, and Seattle. Fourth, we release manuscript-facing claim, table, ablation, and evidence-lane artifacts that make the reported claims auditable.

ORCID(s):

Table 1
PeMS7_228 Stage12 held-out GLS/MAP MAE. Lower is better. Paired p values compare TRACE-SL with the listed baseline on the same ten splits.

Layout	10%	20%	30%
TRACE-SL	**3.590**	**3.355**	**3.084**
Best random by validation	3.713	3.450	3.247
Random mean	3.833	3.575	3.404
Top variance	3.730	3.428	3.200
Greedy A-trace	3.686	3.465	3.351
Observability proxy	4.001	3.924	3.887
Graph sampling	4.133	3.992	3.805
QR/POD	3.977	3.784	3.447
Multistart swap	3.595	3.397	3.204
p vs. best random	8.8e-05	0.0054	0.0005
p vs. random mean	9.5e-08	1.2e-06	7.6e-08
p vs. top variance	0.0004	0.0026	0.0042

The rest of the paper is organized as follows. Section 2 positions TRACE-SL relative to traffic count location, sensor selection, graph sampling, and traffic estimation. Section 3 defines the reconstruction-aware layout problem. Section 4 presents TRACE-SL and its scoped theoretical statements. Sections 5 and 6 report main evidence, ablations, and robustness routing. Section 7 discusses limitations, and Section 8 concludes.

2. Related work

Traffic count, sensor location, and observability. Transportation sensor location has usually been studied as an infrastructure design problem for OD estimation, link-flow inference, path reconstruction, or travel-time monitoring. Early count-location work derived OD covering, flow-fraction, flow-interception, and link-independence rules for OD matrix estimation (Yang and Zhou, 1998). Later reviews and network sensor location models distinguish flow-estimation and flow-observability objectives, including bilevel formulations, basis-link selection, graph-theoretic link inference, and heterogeneous active/passive sensors (Gentili and Mirchandani, 2011; Hu et al., 2009; He, 2013; Fu, Zhu, Ling, Ma and Huang, 2016; Fu, Zhu and Ma, 2017). These papers provide the closest transportation-system design lineage for TRACE-SL. The difference is the target quantity: TRACE-SL does not seek OD identifiability, complete link-flow observability, or path reconstruction. It asks which fixed sparse sensor set makes the hidden traffic state recoverable under a transparent reconstruction estimator and a held-out validation protocol.

Traffic state estimation from sparse observations. Traffic state estimation treats the network state itself as the object of inference. Classical approaches include filtering, model-based data assimilation, and reconstruction from partial or aggregated measurements; surveys emphasize that traffic variables are often noisy and only partially observed (Seo et al., 2017). Signal reconstruction methods have also been used to recover traffic states from aggregated or limited measurements (Coric et al., 2012). This literature motivates hidden-network reconstruction as an operational objective, but it often assumes the sensing pattern is given. TRACE-SL moves one layer upstream: the estimator remains transparent and intentionally simple, while the layout is optimized for the downstream reconstruction task.

Sensor selection and optimal experimental design. The broader optimization and statistical learning literature studies subset selection for estimation. Convex and greedy sensor-selection methods optimize A-, D-, or E-type design criteria under linear models (Joshi and Boyd, 2009). Gaussian-process placement maximizes information gain and obtains approximation guarantees from submodularity (Krause, Singh and Guestrin, 2008b; Krause, McMahan, Guestrin and Gupta, 2008a). TRACE-SL borrows the idea that posterior uncertainty can guide placement, but it does not inherit those approximation guarantees: traffic data are non-Gaussian, validation loss is MAE-oriented, and the final selector combines multiple certificate, coverage, and reconstruction-score channels. The paper therefore treats posterior trace and log determinant as scoped certificates inside a validation-calibrated traffic design workflow.

Graph sampling, POD, and sparse reconstruction. Graph signal processing studies when graph-supported signals can be reconstructed from selected vertices, with sampling-set criteria tied to spectral structure (Chen, Varma, Sandryhaila and Kovacevic, 2015; Anis, Gadde and Ortega, 2016). Data-driven sparse sensor placement similarly exploits low-dimensional modal structure to reconstruct high-dimensional states from few measurements (Manohar, Brunton, Kutz and Brunton, 2018). TRACE-SL includes graph-sampling and QR/POD-style baselines because they are natural competitors for recoverability-oriented sensing. Its methodological contribution is not a new graph sampling theorem; it is a traffic deployment protocol that compares those generic reconstruction criteria against validation-calibrated layouts on hidden-network traffic state recovery.

Learning-based traffic forecasting. Deep spatiotemporal traffic models, including diffusion recurrent and graph convolutional forecasting models, have improved prediction accuracy on fixed sensor graphs (Li et al., 2018; Yu et al., 2018). Such estimators are complementary to TRACE-SL. They answer how to forecast or impute once a sensor graph exists; TRACE-SL answers which sparse graph should be observed if the downstream goal is transparent hidden-state reconstruction. This separation keeps the sensor layout claim from being conflated with a black-box estimator benchmark.

3. Reconstruction-aware sensor layout

Let $G = (\mathcal{V}, E)$ denote a traffic network with candidate sensing locations $\mathcal{V}$. At each time t, the traffic state is $x_t \in \mathbb{R}^{|\mathcal{V}|}$. A deployment chooses a fixed sensor set $\mathcal{S} \subseteq \mathcal{V}$ with $|\mathcal{S}| \leq k$ and observes $x_{t,\mathcal{S}}$, while the hidden complement $\mathcal{H} = \mathcal{V} \setminus \mathcal{S}$ must be reconstructed. The design objective is not to recover an OD matrix or guarantee full observability; it is to choose $\mathcal{S}$ so that a transparent reconstruction rule $\hat{x}_{t,\mathcal{H}} = f_{\mathcal{S}}(x_{t,\mathcal{S}})$ has low held-out error on hidden components.

Definition 1 (Budgeted hidden-network reconstruction design)**.** *Given a candidate set $\mathcal{V}$, a sensor budget k, a training split $\mathcal{D}_{\mathrm{tr}}$, selector-validation split $\mathcal{D}_{\mathrm{sel}}$, tuner-validation split $\mathcal{D}_{\mathrm{tun}}$, test split $\mathcal{D}_{\mathrm{te}}$, and a transparent reconstruction family $\{f_{\mathcal{S}} : |\mathcal{S}| \leq k\}$, the design problem is to output a fixed layout $\hat{\mathcal{S}}$ before test evaluation so that hidden-component deployment risk*

$$R(\mathcal{S}) = \mathbb{E}\left[\ell\left(x_{\mathcal{H}}, f_{\mathcal{S}}(x_{\mathcal{S}})\right)\right], \qquad \mathcal{H} = \mathcal{V} \setminus \mathcal{S},$$

is small for a reconstruction loss ℓ. The empirical paper metric uses hidden-component MAE on $\mathcal{D}_{\mathrm{te}}$ after $\hat{\mathcal{S}}$ and all layout-selection rules are fixed.

The experiments use transparent GLS/MAP and graph-signal reconstruction models. Training data estimate reconstruction ingredients such as means, covariances or precisions, graph operators, shrinkage levels, and regularization. Selector-validation data choose the best layout within each candidate scoring rule. Tuner-validation data choose the scoring rule or score vector. Test data are used only after the layout rule is fixed. This split discipline prevents validation reconstruction error from being treated as final deployment evidence.

For a candidate layout $\mathcal{S}$, the primary test metric is hidden-component MAE,

$$\mathrm{MAE}(\mathcal{S}) = \frac{1}{|\mathcal{T}_{\mathrm{test}}||\mathcal{H}|} \sum_{t \in \mathcal{T}_{\mathrm{test}}} \|x_{t,\mathcal{H}} - \hat{x}_{t,\mathcal{H}}\|_1.$$

The paper-source artifacts also report RMSE, MAPE where available, posterior trace, condition number, log determinant, scenario CVaR trace, coverage, and validation diagnostics. The central claim is empirical and scoped: layouts selected by TRACE-SL improve reconstruction in the tested settings and under the stated transparent reconstruction protocol.

This formulation deliberately keeps the target narrower than complete network observability. A layout may fail to identify every latent path or OD flow and still be useful for hidden-state recovery if it improves the reconstruction loss of the traffic state variables actually used by the downstream estimator. Conversely, a layout with strong topological coverage can be weak under the held-out reconstruction metric if the observed locations are redundant for the temporal-spatial variation in the hidden complement.

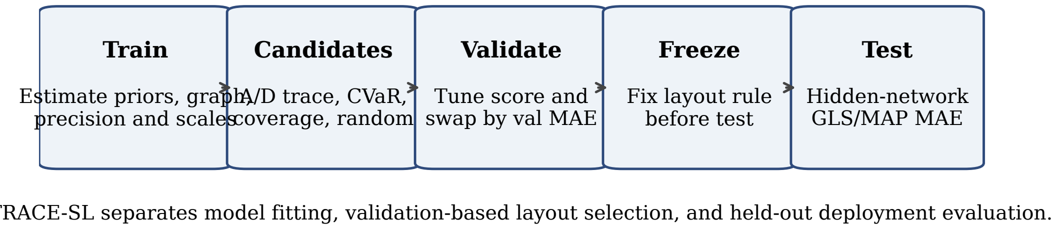

Figure 1: TRACE-SL workflow. The design protocol separates training-derived reconstruction ingredients, validation-based layout selection, and held-out deployment evaluation.

4. TRACE-SL method and scoped theory

TRACE-SL is a reconstruction-aware candidate-selection and refinement protocol. It is not a single closed-form score. The method is organized around three decisions that are common in deployment studies: which candidate layouts to consider, which score vector to trust on validation data, and whether a local exchange improves actual reconstruction rather than only a proxy certificate.

4.1. Transparent reconstruction layer

For a fixed layout $\mathcal{S}$, the GLS/MAP reconstruction layer estimates training means and covariances and reconstructs hidden components from observed components. In the linear-Gaussian version used to motivate the certificates, let

$$x \sim \mathcal{N}(\mu, \Sigma), \qquad y_{\mathcal{S}} = H_{\mathcal{S}} x + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, R_{\mathcal{S}}),$$

where $H_{\mathcal{S}}$ selects observed components and $R_{\mathcal{S}}$ is observation noise. The posterior mean is the transparent reconstruction rule and the hidden posterior covariance is

$$\Sigma_{\mathcal{H}|\mathcal{S}} = \Sigma_{\mathcal{H}\mathcal{H}} - \Sigma_{\mathcal{H}\mathcal{S}} \left(\Sigma_{\mathcal{S}\mathcal{S}} + R_{\mathcal{S}}\right)^{-1} \Sigma_{\mathcal{S}\mathcal{H}}.$$

Graph-signal reconstruction uses the same layout interface but replaces the covariance-conditioned estimator with graph-regularized recovery. TRACE-SL can therefore compare covariance, graph, POD, and validation losses without changing the held-out metric.

4.2. Candidate generation and validation-calibrated selection

The candidate pool is intentionally heterogeneous. Posterior A-trace and D-logdet represent classical uncertainty certificates; scenario-average and scenario-CVaR trace expose average-case and tail-risk variants; top variance and degree are simple reviewer-facing heuristics; graph sampling and QR/POD represent graph-signal and modal reconstruction baselines; random and validation-selected random isolate the value of validation selection from certificate-guided candidate generation. The final TRACE-SL layout is selected by reconstruction loss on validation data rather than by assuming that any single proxy score is universally correct.

4.3. Formal scoped properties

Proposition 1 (Posterior trace identity)**.** *Under the linear-Gaussian model above, the posterior mean* $\mathbb{E}[x_{\mathcal{H}} \mid y_{\mathcal{S}}]$ *minimizes conditional squared reconstruction error among measurable estimators of* $x_{\mathcal{H}}$*, and*

$$\mathbb{E}\left[\|x_{\mathcal{H}} - \mathbb{E}[x_{\mathcal{H}} \mid y_{\mathcal{S}}]\|_2^2 \mid \mathcal{S}\right] = \mathrm{tr}(\Sigma_{\mathcal{H}|\mathcal{S}}).$$

Proof. For squared loss, the conditional expectation is the Bayes estimator. Conditional on $y_{\mathcal{S}}$, the residual $x_{\mathcal{H}} - \mathbb{E}[x_{\mathcal{H}} \mid y_{\mathcal{S}}]$ has covariance $\Sigma_{\mathcal{H}|\mathcal{S}}$. Taking the trace of the residual second moment gives the conditional expected squared Euclidean error. The statement is conditional on the fixed prior and observation model; it does not imply an MAE guarantee for non-Gaussian traffic data. □

Table 2
Algorithm 1: TRACE-SL candidate generation and validation-calibrated layout selection.

Input	traffic graph $G = (\mathcal{V}, E)$; budget k; train, selector-validation, tuner-validation, and test splits; candidate generator family $\mathcal{G}$; score vectors Λ; transparent reconstruction family $f_{\mathcal{S}}$; swap limits.
1	Fit reconstruction ingredients on training data: means, covariance/precision or graph operators, shrinkage, and regularization.
2	For each generator $g \in \mathcal{G}$, build candidate layouts using posterior A-trace, D-logdet, scenario-average trace, scenario-CVaR trace, top variance, coverage, degree, graph sampling, QR/POD, and random candidate families.
3	For each score vector $\lambda \in \Lambda$, score candidates using certificate, coverage, conditioning, and validation diagnostics; choose the best selector-validation layout for that vector.
4	Evaluate the selected layout for each λ on tuner-validation reconstruction MAE and choose $\lambda^\star$ with the lowest tuner-validation loss.
5	Starting from the chosen layout and strong internal candidates, run Algorithm 3 for validation-aware one-swap refinement.
6	Freeze the resulting layout $\hat{\mathcal{S}}$ and evaluate hidden-component MAE/RMSE/MAPE on the held-out test split only.

Table 3
Algorithm 2: validation-aware one-swap refinement.

Input	initial layout $\mathcal{S}_0$; add pool $\mathcal{A} \subseteq \mathcal{V} \setminus \mathcal{S}_0$; remove pool $\mathcal{R} \subseteq \mathcal{S}_0$; validation loss $\hat{R}_{\text{tun}}(\mathcal{S})$; maximum starts and iterations.
1	Set $\mathcal{S} \leftarrow \mathcal{S}_0$ and compute $\hat{R}_{\text{tun}}(\mathcal{S})$.
2	Enumerate feasible exchanges (r, a) with $r \in \mathcal{R} \cap \mathcal{S}$ and $a \in \mathcal{A} \setminus \mathcal{S}$, preserving $\lvert\mathcal{S}\rvert = k$.
3	Evaluate $\hat{R}_{\text{tun}}((\mathcal{S} \setminus \{r\}) \cup \{a\})$ for the configured neighborhood.
4	Accept the best strict validation improvement; update $\mathcal{S}$, $\mathcal{A}$, and $\mathcal{R}$ if the implementation refreshes pools.
5	Stop when no evaluated exchange strictly improves validation loss or when iteration limits are reached. Return $\mathcal{S}$.

Proposition 2 (Posterior covariance monotonicity for fixed targets)**.** *Fix a target set $\mathcal{Q} \subseteq \mathcal{V}$ whose state is to be reconstructed, and compare two observation sets $\mathcal{S} \subseteq \mathcal{S}'$ disjoint from $\mathcal{Q}$ under the same prior and observation-noise model. Then*

$$\Sigma_{\mathcal{Q}|\mathcal{S}'} \preceq \Sigma_{\mathcal{Q}|\mathcal{S}}, \qquad \operatorname{tr}(\Sigma_{\mathcal{Q}|\mathcal{S}'}) \leq \operatorname{tr}(\Sigma_{\mathcal{Q}|\mathcal{S}}).$$

Proof. Conditioning on $\mathcal{S}'$ is equivalent to conditioning first on $\mathcal{S}$ and then on the additional observations $\mathcal{S}' \setminus \mathcal{S}$. The Gaussian conditioning update subtracts a positive semidefinite Schur-complement term from the previous posterior covariance. Trace monotonicity follows from the Loewner-order relation. The fixed-target condition is needed because the hidden complement changes when sensors are added. □

Proposition 3 (Finite-candidate validation selection bound)**.** *Let C be a finite candidate-layout class evaluated on an independent validation split of size n_{v}. Suppose the validation losses $\ell(x_{\mathcal{H}}, f_{\mathcal{S}}(x_{\mathcal{S}}))$ are bounded in $[0, B]$ for every $\mathcal{S} \in C$. Then, for any $\epsilon > 0$,*

$$\Pr\left(\sup_{\mathcal{S}\in C}\left|R(\mathcal{S}) - \hat{R}_{\text{v}}(\mathcal{S})\right| > \epsilon\right) \leq 2|C|\exp\left(-\frac{2n_{\text{v}}\epsilon^2}{B^2}\right).$$

Proof. For a fixed layout, Hoeffding's inequality controls the deviation between validation loss and deployment risk. A union bound over the finite candidate class gives the displayed inequality. For sub-exponential traffic losses the same role is played by the corresponding Bernstein-type bound with different constants. The proposition supports validation-calibrated selection as a statistically meaningful model-selection step; it is not a test-set performance theorem. □

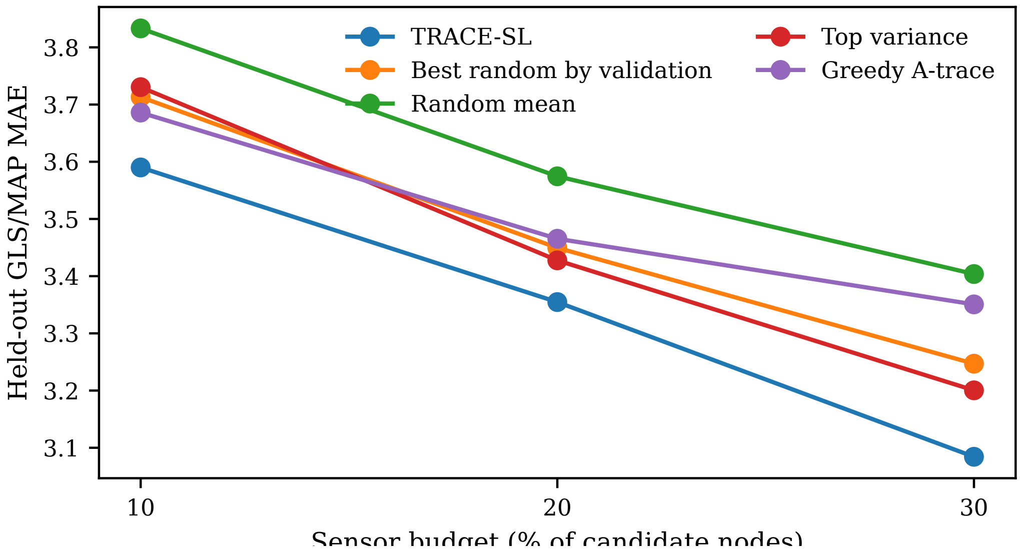

Figure 2: Held-out PeMS7_228 GLS/MAP reconstruction MAE across sensor budgets. TRACE-SL refers to the validation-aware selected layout.

Proposition 4 (Finite termination and evaluated-neighborhood local optimality)**.** *For fixed validation data, deterministic validation loss, finite add/remove pools, and finite iteration limits, Algorithm 3 terminates. If it stops because no evaluated remove/add exchange strictly improves validation loss, the returned layout is locally optimal with respect to the evaluated one-swap neighborhood.*

Proof. The configured neighborhood is finite and each accepted step strictly decreases the deterministic validation loss among evaluated layouts. With finite iteration limits termination is immediate; without repeated strict improvements among a finite set also cannot continue indefinitely. At the no-improvement stopping condition, every evaluated one-swap neighbor has validation loss at least as large as the returned layout. The statement says nothing about unevaluated swaps, larger neighborhoods, or global optimality. □

Remark 1 (Complexity and non-claims)**.** *The workload decomposes into candidate generation, candidate scoring, reconstruction evaluation, and swap evaluation. If M candidates are evaluated, $|\Lambda|$ score vectors are tuned, n_v validation time points are used, and each reconstruction evaluation costs $C_{rec}(k, |\mathcal{V}|)$, the validation-scoring component scales as $O(M|\Lambda|n_v C_{rec})$. Swap refinement adds the number of evaluated remove/add exchanges times the same reconstruction cost. TRACE-SL therefore reports launcher and manifest artifacts rather than claiming polynomial-time optimal placement certification, submodularity, approximation ratios, or universally monotone empirical MAE.*

5. Experimental design and main results

5.1. Datasets, splits, and metrics

The Stage12 experiments evaluate three traffic networks: PeMS7_228, PeMS7_1026, and Seattle. Each network is evaluated at 10%, 20%, and 30% sensor budgets, with budget counts obtained by applying the fraction to the candidate sensor set and then using the run manifest's deterministic rounding rule. Each reported aggregate uses ten held-out splits. Training data fit reconstruction ingredients; selector-validation data choose candidates within a score vector; tuner-validation data select score vectors and accept swap refinements; test data are used only for final reporting.

The primary metric is hidden-component MAE under the transparent GLS/MAP reconstruction protocol. RMSE and MAPE are retained in the paper-source contracts when available, but the main claim is written around MAE because it is directly interpretable in the same scale as the traffic state variable and is less dominated by large residuals than squared loss. All reported paired deltas compare TRACE-SL with a baseline on the same split index. The tables

Table 4
Scoped theoretical statements used in the manuscript. Each statement is intentionally bounded by its assumptions and non-claim boundary.

Statement	Scoped content	Non-claim boundary
budgeted hidden network reconstruction	Choose a fixed sensor set S with |S| <= k from observed candidates; learn reconstruction ingredients on training splits, tune layout selection on validation splits, and report final reconstruction only on held-out test splits after the selection rule is fixed.	does not assert global optimality or guaranteed MAE improvement
posterior trace squared error identity	Under the stated linear-Gaussian squared-error model, the posterior covariance trace over hidden components equals the conditional expected squared reconstruction error up to the fixed observation/noise model used by the estimator.	does not convert posterior trace into an MAE guarantee for non-Gaussian traffic data
posterior covariance monotonicity	For the same linear-Gaussian model, conditioning on an additional sensor cannot increase the posterior covariance in positive-semidefinite order, so posterior trace is nonincreasing as sensors are added.	does not imply submodularity, approximation ratio, or monotone empirical MAE on every split
validation aware one swap local optimality	A validation-aware swap output is locally optimal with respect to the evaluated one-swap neighborhood when no tested remove/add exchange from the configured candidate pool strictly improves validation reconstruction loss.	does not assert global optimality outside the evaluated one-swap neighborhood
finite candidate validation selection	For a finite candidate class and independent validation split, empirical validation risk uniformly concentrates around deployment risk at a rate controlled by $\sqrt{\log \|C\|/n_v}$ under bounded or sub-exponential loss assumptions.	does not convert validation ranking into a held-out test guarantee or universal deployment guarantee
rcss candidate search evaluation complexity	RCSS runtime can be stated as candidate generation plus candidate scoring plus validation-aware swap evaluation, scaling with the configured candidate counts, budgets, validation windows, swap starts, swap iterations, and add/remove pool sizes recorded by launchers and manifests.	does not claim polynomial-time optimal placement certification or hardware-independent runtime guarantees

report ten-split means and, where used for claims, paired t tests, Wilcoxon signed-rank tests, confidence intervals, and win counts in the source artifacts.

5.2. Baselines

The baseline portfolio is designed to separate mechanism from performance. Random mean averages unselected random layouts, while best random by validation tests whether validation selection alone is enough. Top variance selects high-variance locations. Greedy A-trace and greedy D-logdet are certificate-only OED-style layouts. Observability proxy and degree represent transportation/topological heuristics. Graph sampling and QR/POD represent graph-signal and low-rank modal placement. Scenario-average and scenario-CVaR variants test risk-weighted posterior certificates. RCSS-selected layouts test score-based portfolio selection before local refinement. Multistart validation swap is retained as a strong internal comparator rather than hidden as an implementation detail.

5.3. PeMS7_228 main evidence

Table 1 reports the main PeMS7_228 evidence. TRACE-SL reaches mean held-out GLS/MAP MAE of 3.590, 3.355, and 3.084 at 10%, 20%, and 30% budgets. Against validation-selected random layouts, the paired mean deltas

Table 5
External Stage12 evidence with baseline context. Values are ten-split held-out GLS/MAP MAE; lower is better.

Dataset	Budget	TRACE-SL	Best rand.	Greedy A	Top var.	QR/POD	Graph samp.	Obs.	Boundary note
PeMS7_1026	10%	3.767	3.879	3.739	4.026	4.085	4.853	4.484	Greedy A and swap-from-A are slightly lower; TRACE-SL still improves over random, variance, POD, graph, and observability baselines.
PeMS7_1026	20%	3.347	3.560	3.351	3.724	3.545	4.561	4.348	TRACE-SL is close to RCSS/A-trace and better than the reviewer-facing baselines.
PeMS7_1026	30%	3.074	3.332	3.096	3.430	3.168	4.475	4.183	TRACE-SL is the best row among the listed baselines and strong internal variants.
Seattle	10%	3.101	3.218	3.123	3.551	3.465	3.939	3.573	TRACE-SL is effectively tied with RCSS and multistart; no low-budget dominance claim is made.
Seattle	20%	2.828	2.944	2.904	3.101	3.066	3.760	3.427	TRACE-SL improves over key reviewer-facing baselines.
Seattle	30%	2.624	2.739	2.775	2.834	2.807	3.571	3.410	TRACE-SL improves over key reviewer-facing baselines and strong internal comparators.

are -0.123, -0.095, and -0.163 with paired p values 8.8e-05, 0.0054, and 0.0005. Against random mean layouts, the paired mean deltas are -0.243, -0.220, and -0.320.

The low-budget comparison with the multistart swap baseline is deliberately reported as a caveat. At 10%, TRACE-SL and multistart validation swap are statistically close (3.590 versus 3.595, paired $p = 0.812$), and the evidence does not justify a claim that one selected layout dominates every strong internal variant at every budget. The paper-level claim is therefore about reconstruction-aware layout design and the TRACE-SL portfolio, not about a universal winner on every row.

5.4. External-network evidence

Table 5 expands the external evidence beyond TRACE-SL-only MAE. PeMS7_1026 and Seattle both have ten tracked Stage12 splits and include the same major baseline families. The table makes two boundary cases visible. On PeMS7_1026 at 10%, greedy A-trace and swap-from-A-trace are slightly better than the final TRACE-SL selection, while TRACE-SL remains better than random, top-variance, QR/POD, graph-sampling, and observability baselines. On Seattle at 10%, TRACE-SL is effectively tied with RCSS and multistart swap. At higher budgets, TRACE-SL improves over the listed reviewer-facing baselines on both external networks.

This evidence supports a multi-network empirical claim, not a universal generalization theorem. The same protocol transfers across the tested networks, but the external rows also show why the manuscript avoids saying that the final selector is globally optimal or uniformly dominant over every internal variant.

6. Ablations and robustness routing

The ablation contract separates the roles of random candidate pools, validation selection, certificate-guided candidate generation, RCSS selection, validation-aware local refinement, and multistart swap search. Table 6 shows the PeMS7_228 component comparison at 10%, 20%, and 30% budgets. At 10%, TRACE-SL improves over random mean, validation-selected random, and greedy A-trace, but the multistart swap comparator is statistically indistinguishable. At 20% and 30%, local refinement gives clearer gains over the reviewer-facing baselines and over the multistart row at 30%.

The ablation should be interpreted at the component level rather than as a kitchen-sink heuristic. Greedy A-trace and scenario risk variants are certificate-only candidates. Validation-selected random isolates validation selection without certificate-guided candidate generation. RCSS-selected layouts isolate score-based selection before local refinement. Validation-swap selected layouts add the local neighborhood search. Multistart swap is retained as a serious comparator because it is close at low budget and strong enough to prevent an overbroad dominance claim.

Robustness evidence is routed as stress-test evidence. The project includes perturbation artifacts for sensor failure, observation noise, missingness, heterogeneous costs, temporal shift, and candidate-pool sensitivity. These tests support bounded robustness discussion in the tested settings. They do not support the phrase globally robust, and they do not replace held-out Stage12 evidence for main performance claims.

Table 6
PeMS7_228 component ablation across budgets. Values are ten-split held-out GLS/MAP MAE; lower is better.

Layout component	10%	20%	30%
Random mean	3.833	3.575	3.404
Best random by validation	3.713	3.450	3.247
Greedy A-trace	3.686	3.465	3.351
RCSS selected	3.605	3.393	3.183
TRACE-SL	**3.590**	**3.355**	**3.084**
Multistart swap	3.595	3.397	3.204

7. Discussion and limitations

TRACE-SL is useful because it aligns a costly infrastructure decision with the downstream inverse problem. A high-flow or high-variance sensor can be attractive, but a reconstruction-aware layout may instead choose sensors that reduce hidden-state uncertainty and improve validation recoverability. The method is intentionally transparent: its certificate terms, validation losses, selected layouts, and held-out outcomes can be inspected.

Several limitations define the claim boundary. First, posterior trace is a squared-error uncertainty certificate under scoped GLS/MAP assumptions; it is not a theorem that MAE must improve on non-Gaussian traffic data. Second, validation-aware swap search is local to the evaluated one-swap neighborhood and candidate pool. Third, PeMS7_1026 and Seattle support multi-network empirical evidence, not universal cross-network generalization. Fourth, robustness results are stress tests rather than formal certification against all perturbations.

These limitations are methodological guardrails rather than defects. They make the manuscript falsifiable: a reviewer can trace each claim to a table, theorem statement, or explicit caveat. Future work should study stronger stochastic or bilevel formulations, approximation properties under traffic-specific covariance structure, and deployment studies with real maintenance and sensor-failure costs.

The submission package therefore treats auditability as part of the method. Paper-visible numbers are generated from machine-readable Stage12 artifacts, the main caveats are repeated where they affect interpretation, and the source separates manuscript prose from raw traffic datasets. This structure is intended to make later revisions local: if a reviewer questions a number, baseline, or scope statement, the corresponding row in the claim, ablation, or external-evidence contract identifies the evidence lane to inspect.

8. Conclusion

TRACE-SL reframes sparse traffic sensor deployment as a recoverability-driven inverse-design problem. Instead of ranking locations by coverage, variance, or topology alone, it selects a fixed sensor set by how well the hidden network state can be reconstructed under transparent estimators and validation-calibrated layout rules.

The method combines posterior and scenario certificates with validation selection and local refinement, and evaluates layouts only after the selection rule is fixed. Across PeMS7_228, PeMS7_1026, and Seattle Stage12 evidence, the resulting layouts consistently improve over standard random, variance, observability, graph-sampling, and POD-style baselines in the tested settings, while explicitly exposing low-budget cases where internal A-trace, RCSS, or multistart variants are competitive. The final contribution is an auditable bridge between traffic network design, transparent inverse problems, and empirical sensor deployment evidence.

CRediT authorship contribution statement

Anonymous for review.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

The manuscript reports curated aggregate artifacts and generation scripts under `TRC-23-02333/trace_sl_results/paper_sources/`. Raw traffic datasets are local inputs and are not distributed with the paper source; the source package records the expected local file layout, launch scripts, aggregate CSV/JSON contracts, and audit artifacts used to regenerate paper-visible tables. The reported numbers should therefore be read as reproducible from the curated artifacts, while full raw-data reproduction requires obtaining the underlying traffic datasets separately.

References

Anis, A., Gadde, A., Ortega, A., 2016. Efficient sampling set selection for bandlimited graph signals using graph spectral proxies. IEEE Transactions on Signal Processing 64, 3775–3789. doi:10.1109/TSP.2016.2546233.

Chen, S., Varma, R., Sandryhaila, A., Kovacevic, J., 2015. Discrete signal processing on graphs: Sampling theory. IEEE Transactions on Signal Processing 63, 6510–6523. doi:10.1109/TSP.2015.2469645.

Coric, V., Djuric, N., Vucetic, S., 2012. Traffic state estimation from aggregated measurements with signal reconstruction techniques. Transportation Research Record: Journal of the Transportation Research Board 2315, 121–130. doi:10.3141/2315-13.

Fu, C., Zhu, N., Ling, S., Ma, S., Huang, Y., 2016. Heterogeneous sensor location model for path reconstruction. Transportation Research Part B: Methodological 91, 77–97. doi:10.1016/j.trb.2016.04.013.

Fu, C., Zhu, N., Ma, S., 2017. A stochastic program approach for path reconstruction oriented sensor location model. Transportation Research Part B: Methodological 102, 210–237. doi:10.1016/j.trb.2017.05.013.

Gentili, M., Mirchandani, P.B., 2011. Survey of models to locate sensors to estimate traffic flows. Transportation Research Record: Journal of the Transportation Research Board 2243, 108–116. doi:10.3141/2243-13.

He, S.X., 2013. A graphical approach to identify sensor locations for link flow inference. Transportation Research Part B: Methodological 51, 65–76. doi:10.1016/j.trb.2013.02.006.

Hu, S.R., Peeta, S., Chu, C.H., 2009. Identification of vehicle sensor locations for link-based network traffic applications. Transportation Research Part B: Methodological 43, 873–894. doi:10.1016/j.trb.2009.02.008.

Joshi, S., Boyd, S., 2009. Sensor selection via convex optimization. IEEE Transactions on Signal Processing 57, 451–462. doi:10.1109/TSP.2008.2007095.

Krause, A., McMahan, H.B., Guestrin, C., Gupta, A., 2008a. Robust submodular observation selection. Journal of Machine Learning Research 9, 2761–2801. URL: https://www.jmlr.org/papers/v9/krause08b.html.

Krause, A., Singh, A., Guestrin, C., 2008b. Near-optimal sensor placements in gaussian processes: Theory, efficient algorithms and empirical studies. Journal of Machine Learning Research 9, 235–284. URL: https://www.jmlr.org/papers/v9/krause08a.html.

Li, Y., Yu, R., Shahabi, C., Liu, Y., 2018. Diffusion convolutional recurrent neural network: Data-driven traffic forecasting, in: International Conference on Learning Representations. URL: https://openreview.net/forum?id=SJiHXGWAZ.

Manohar, K., Brunton, B.W., Kutz, J.N., Brunton, S.L., 2018. Data-driven sparse sensor placement for reconstruction: Demonstrating the benefits of exploiting known patterns. IEEE Control Systems Magazine 38, 63–86. doi:10.1109/MCS.2018.2810460.

Seo, T., Bayen, A.M., Kusakabe, T., Asakura, Y., 2017. Traffic state estimation on highway: A comprehensive survey. Annual Reviews in Control 43, 128–151. doi:10.1016/j.arcontrol.2017.03.005.

Yang, H., Zhou, J., 1998. Optimal traffic counting locations for origin-destination matrix estimation. Transportation Research Part B: Methodological 32, 109–126. doi:10.1016/S0191-2615(97)00016-7.

Yu, B., Yin, H., Zhu, Z., 2018. Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting, in: Proceedings of the Twenty-Seventh International Joint Conference on Artificial Intelligence, pp. 3634–3640. doi:10.24963/ijcai.2018/505.

A. Additional audit notes

The source package is generated from `NARRATIVE_REPORT.md` and CSV/JSON paper-source artifacts under `TRC-23-02333/trace_sl_results/paper_sources/`. Raw datasets are not embedded in the paper source. Numeric claims in the main text should be rechecked by `paper-claim-audit` after every editing round.

B. Reproducibility route

The paper-facing evidence is organized as contracts rather than freehand manuscript numbers. The main PeMS7_228 table is generated from `core_performance_table.csv` and `paired_delta_table.csv`. External-network comparisons are routed through `external_evidence_contract.csv`. Component claims use `ablation_contract.csv`; scoped theoretical statements use `theory_statement_contract.csv`. Each contract records source directories, source CSV names, split counts, evidence status, and caveat tags. This structure is intended to make reviewer-driven changes local: if a table row or caveat changes, the corresponding contract row identifies the generating experiment lane.

Full reproduction requires obtaining the underlying PeMS and Seattle traffic datasets and placing them in the local dataset layout expected by the experiment scripts. The paper source itself distributes aggregate artifacts and launch metadata, not raw traffic data.

C. Proof sketches for scoped statements

The main text contains the formal proposition statements used for the manuscript claims. This appendix records the proof obligations and non-claim boundaries in prose.

For the posterior trace identity, the proof follows the standard conditional Gaussian calculation: under a fixed linear observation model and squared-error loss, the conditional mean minimizes expected squared error and the residual covariance over hidden components is the posterior covariance. Taking the trace gives the expected squared Euclidean reconstruction error over hidden components. This statement is tied to squared loss and the fitted linear-Gaussian model.

For monotonicity, adding a sensor corresponds to conditioning on an additional linear observation under the same prior/noise model. The Schur complement update subtracts a positive semidefinite term from the previous posterior covariance, hence the posterior covariance is nonincreasing in Loewner order and its trace is nonincreasing. The comparison must use a fixed reconstruction target; otherwise the hidden complement changes with the layout.

For validation selection, the proof uses a concentration inequality for each fixed candidate layout and then unions over the finite candidate class. The bound justifies validation-calibrated selection as a finite-class model-selection step, not as a post-selection guarantee on the held-out test split.

For validation-aware swap local optimality, the algorithm terminates after evaluating the configured finite neighborhood. If no evaluated remove/add exchange strictly reduces validation reconstruction loss, the returned layout is locally optimal with respect to that evaluated neighborhood. The statement is not about swaps outside the configured add/remove pools or about global optimality.